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Final Project

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Introduction & Background

Municipal bond markets exist in various parts of the world. However, the American municipal bond market is considered the most developed and vibrant of all the markets. The City of New York issued the first municipal bond for a canal project in 1812. Several bonds were issued to fund public infrastructure projects, such as railroads and public schools (Municipal Bond, 2023). Today, the value of bonds issued in the municipal bond market is around $4 trillion.

To understand municipal bonds, we must understand a plain vanilla bond. A plain vanilla bond essentially represents a loan made to the issuer by the investors. Here, the issuer must pay the interest plus the principal regularly to the investor providing funding for the loan. Take a simplified example of a 10-year $100 bond with a 2% annual coupon, which means that in year 0, i.e., in the beginning, the investor hands over $100 to the issuer as a loan, and from year one till year ten, the issuer must make periodic payments of $2 (2% of $100) every year and the principal repayment of $100 in year 10. Thus, the investor keeps getting yearly interest and recovers the principal in the final year. This is generally the payment structure of a plain vanilla bond.

Municipal Bonds, also known as muni bonds, operate on a similar logic. States, cities, and counties generally issue them and are a vital source of financing for these governmental entities. Furthermore, every bond issuer must issue an offering document stating numerous details about the bond, such as financial information, bond yields, payment schedules, revenue sources, etc. Therefore, these offering documents are analogous to a prospectus but are more formal, and the issuers are legally responsible for providing the right information to the bond buyer. The issuers may disclose more information to the buyers, such as various risk factors associated with the bonds.

Some of these factors include but are not limited to credit risk, default risk, inflation, interest rates, and so on. For example, if the bond's credit risk increases, the bond becomes more risky, and investors incur a loss. However, in this paper, our focus will not be on economic variables because the fact that governmental entities issue municipal bonds exposes these bonds to many political risk factors that affect the yields of these bonds. These political risk factors could be elections, corruption, political misalignment, and so on. Therefore, in our study, we attempt to understand the representation of political risk in the municipal bond offering documents.

Literature Review

Regarding politics and the government, studies have only recently started focusing on how politicians and politics impact municipal bonds. Municipal bonds are issued locally, so local-level elections and the corresponding socioeconomic factors related to that area also impact bond yields. Chen, Li, and Liu (2022) find that municipal bond yields increase by 0.8% for a one-standard-deviation increase in polarization. Abakah (2020) finds that counties with lower Catholic-to-Protestant population ratio are less risky and positively impact bond yields. Furthermore, some critical players in the government, such as governors and their ideological inclinations, may influence fiscal policies, which may affect municipal bonds. Choi et al. (2015) find that credit spreads of municipal bonds increase during gubernatorial elections.

Moreover, Gao, Murphy, and Qi (2019) find that the U.S. gubernatorial elections sharply increase municipal bond yields by 7bps(0.07%) right before the elections. This spike in yields is lower for states with a better economy and more transparency in disclosures and higher for states with closely contested elections. Moreover, the alignment of the counties with the state governor

matters when investors demand higher yields for bonds issued in misaligned counties, especially during closely contested elections (Ren and Zhao 2021).

As far as the offering documents are concerned, using content analysis, Farrell et al. (2023) find that retail investors demand a greater yield if they find the offer documents present information relevant to making investment decisions in a complicated manner. However, not much is discussed in this study related to how political risk is represented in these documents.

These studies on municipal bonds primarily focus on economic variables and their impact on muni bond yields. Moreover, political risk induced by political activities, such as elections and ideological misalignments impacting bond yields, is missing from these studies. There are very few studies done for the US regarding politics regarding offering documents. Such political activities or factors may significantly enhance the riskiness of these bonds. Therefore, studying how political risk impacts the riskiness of bonds mentioned in offering documents will be a good starting point.

Uniqueness

The US municipal bond market is enormous ($4 Trillion), and offering documents are an integral component. Yet very few studies have analyzed offering documents in detail. Thus, our project is unique because we include a large corpus of municipal bond offering documents (n=865) for contentment analysis to feed our supervised and unsupervised algorithms.

Furthermore, most studies use either supervised or unsupervised approaches. We use both approaches to complement each other and uncover the political risks mentioned in these documents. We also develop an original dictionary of political risk words to inform our supervised algorithms.

As discussed previously, municipal bonds may not be free of political risk. The results of the supervised and unsupervised algorithms will help us better understand whether this risk is adequately represented in the documents. These results will have a wide-ranging impact on various market participants such as investors, traders, governments, bond dealers, underwriters, lawyers, etc.

For example, the US municipal bond markets from the supply side consist of public entities such as schools, state and local governments, underwriters, lawyers, etc. The demand side comprises households, mutual funds, institutions, and ETFs (Equity Traded Funds). Greater transparency of disclosures in offering documents could foster an environment of trust between the demand side and supply side entities, leading to more efficient and vibrant capital markets.

Finally, the research design of this study could be replicated to uncover different risks mentioned in these documents. For example, this research design can be used to see if environmental risk emanating from specific projects, such as roads, bridges, airports, etc., is adequately represented in the documents.

Research Question and Hypothesis

This research examines how political risks are disclosed to investors in municipal bond documents and how political volatility at the municipal, state, and federal levels may affect the yields of these bonds. Ideological conflicts and political instability between municipal, state, and federal leaders may influence bond yields as investors adjust their risk premiums in response to perceived political uncertainty.

Primary Research Question: How is political risk information reflected in municipal bond documents? Our test hypothesis is:

H_0: Political risk information is not systematically shared in municipal bond documents

H_1: Political risk information is systematically shared in municipal bond documents

Research Design

To answer our research question, we use supervised and unsupervised approaches that complement each other. We first developed a dictionary to search for words related to political risk in the corpus of 865 documents, followed by descriptive statistics related to the documents. We use various packages such as quanteda, tm, readtext, and so on to perform our analysis.

Then, we used the unsupervised approach using the STM package to uncover different topics of the documents. Using both these approaches, we attempt to uncover to what extent political risk is mentioned in the documents.

Dictionary for Political Risk

Schragger (2006), in his study, finds that cities have a subordinate status in policy, where the mayor's office is merely symbolic and has little authority in budgeting and finance. Some of the words associated with this study were mayor, budget, expenditures, city, revenue, administration, tax, lobby, interest groups, and property.

However, Salanick and Pfeffer (1977) mention that the Mayor is still a power broker between the Government and various interest groups, and he does have some power contrary to the popular belief. This study has the same words associated with the previous one but with another word: bureaucracy.

Moreover, de Benedictis-Kessner and Warshaw (2016) find that Mayor's partisanship matters where a democratic mayor leads to more per-capita spending in the municipality. The mayor has less leeway to change city expenditures but issues bonds to increase the expenditures.

Contrary to previous studies, the author finds that the mayor has a say in city policy. Words: property, partisanship.

Staff (2001) study finds that Governors and Mayors join forces to oppose the funding cuts to the state government by the President. Governors and Mayors lobby Congress to restore the funding back. Some words associated with this study are: cuts and funds.

Kirkland (2021) study finds that if the Mayor was a business owner with a past experience in running a business, then the fiscal policies of the municipalities are less focussed toward redistribution and more towards infrastructure development. Words: fiscal policy, policy area, party, grants.

Condray and Conlan (2019) find that the states have been trying to amend Article V of the American Constitution, which reduces the national government's power. Some of the attempts by the states in the area of finance include forcing the federal government to share some form of revenue with the state. Words: power, separation, federalism, decentralization, authority, district.

Gillette and Skeel (2015), study the municipal bankruptcy law and find that the States themselves should initiate municipal governance reforms. However, political factors and imperatives delay this process. Words: officer, executive, reforms, constitution.

Consolidated Dictionary: mayor, budget, expenditures, city, revenue, administration, Tax, lobby, property, budget, bureaucracy, partisanship, election, spending, cuts, funds, power, separation, federalism, fiscal, executive, reforms, constitution, district, officer, authority, governor, president

Preprocessing

We collected 1,000 primary market offering documents from EMMA Labs. Each of these files were in .pdf format, and on average contained approximately 60-70 pages. Some of these documents were OCR scans of paper originals, while others were direct imports of an electronically formatted type-set original. The upload date range of these documents spanned from 2013 to 2024.

Initially, we performed a variety of pre-processing and formatting steps on our corpus to prepare it for further analysis. This involved first converting the pdfs to .txt files using the 'pdftools' package (Ooms 2024). The end-result of this step left us with 845 documents. Most likely, conversion could not proceed on the remaining 155 .pdfs due to rare encoding and/or OCR artifacts embedded in certain documents. Then, with the base R functions 'iconv' and 'gsub' (R Core Team 2024, 256-257, 272) the .txt files were converted to UTF-8 standard, and extraneous encoded/non-ascii characters were removed. With the use of the '™' package (Feinerer and Hornik 2024), the following steps were performed: Punctuation marks and numbers were also removed, and the entire corpus was converted to lower-case text. In addition, common 'stop-words' were parsed out, as these were considered unlikely to assist in finding underlying patterns in our corpus (Foster et al. 2017, 191). Whitespaces were also eliminated. The corpus was parsed for word roots (stemmed). Finally, extremely sparse stems were then filtered out of our corpus to make for a lighter data object to manipulate.

As noted in Welbers et al. (2017, 245-247, 262), these steps assist in rendering textual data amenable to analysis within the R framework: Specifically, they standardize the components of the text and allow it to be input into objects such as a document-term matrix (DTM) which is easier for quantitative analysis as a DTM is basically a large matrix, which is easily understandable by statistical software such as R.

From the cleaned corpus, a DTM, tibble data-frame, a quanteda corpus object, and a quanteda tokens object were created to use in subsequent analysis steps. A separate non-stemmed DTM, corpus and tokens object were also prepared for use with dictionary-based methods because most of our dictionaries were not structured to search for lexical roots, but entire word(s).

Perhaps the most sensitive decision of our pre-processing procedures involved our choice of which factor-weighting to apply to remove sparse terms from our DTM using the '™' package. We choose to remove terms which had *at least* 10% of their entries within the DTM recorded as empty values (Feinerer and Hornick 2024, 32-33). This choice was based on the trade-off between losing 'resolution' of the corpus by dropping too many terms and keeping enough to be able to make sense of it (Welbers et al. 2017, 253-254, Egerod and Klemmensen 2019, 23), as well as a regard for working with a lighter and therefore more memory efficient DTM. We believe our threshold assisted in grounding subsequent stages of our analysis, although other ranges could of course be justified for other analytical desiderata.

All files and codices can be found in our project GitHub repository (Merrill 2024).

Descriptive Statistics

A quick glance at a listing of the most frequent stems (Benoit et al. 2024b 6-7), in our corpus (Figure 1) reveals phrases that might be expected of financial documents: 'payment', 'asses', and 'date'.

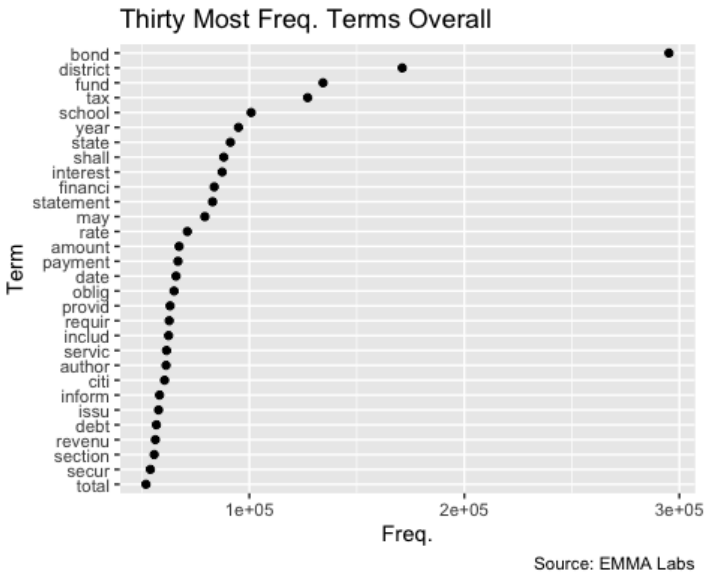


Figure 1

Similar terms like ‘bond’, ‘oblig’, ‘agreement’ would also not be remiss in a corpus composed of bank agreements, accounting spreads, or even financial newsletters. To get an additional visual grasp of our corpus, we charted the most frequent terms that also appeared more than eight-hundred times within a given document (Figure 2). With the ‘quanteda.texplots’ package (Benoit et al. 2024c, 8-9) we also produced a ‘word cloud’ of the most common terms in our corpus (Figure 3):

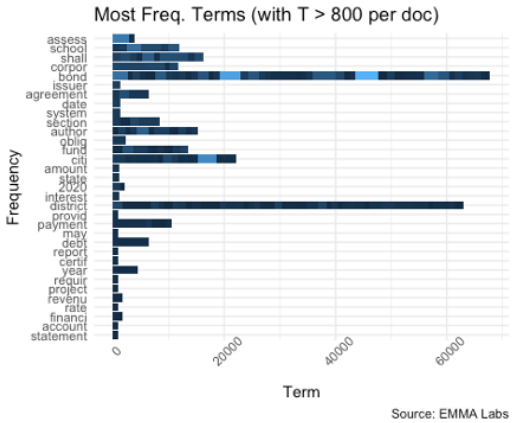


Figure 2

Figure 3

Common financial lexicals like ‘statement’, ‘report’, ‘debt’, and ‘fund’ again seem to predominate. Unsurprisingly, the word-cloud emphasizes how many words such as ‘year’, ‘tax’, ‘district’ and ‘state’ dominate across the range of our corpus. Below (Figure 4) is a chart of the most frequent terms of our corpus by their rank (in the frequency listing) in the overall frequency of all terms (Silge and Robinson 2017, 35):

Figure 4

Though individual terms are not parsed, this graphic does show us that our corpus does *not* evidently obey what is known as 'Zipf's Law' in the aggregate. This phenomenon denotes the tendency of the most frequent terms in a corpus to have higher rank within it, and vice-versa (2017, 34-37); again, a trend not observed within our corpus.

We also calculated the 'term frequency-inverse document frequency' (TF-IDF) of our corpus. This metric allows us to calculate how frequently a word appears, while weighting it against the number of documents it appears in (Foster et al. 2017, 192-193). So, the TF-IDF will be high for words that do not appear in *a lot* of the documents of a corpus. Such a metric poses a relatively balanced way to ascertain words or stems that are important on a document basis within a text-corpus (Silge and Robinson 2017, 44).

We can look at a plot of the twenty highest scoring stems in terms of TF-IDF (2017, 37-39) for our corpus (Figure 5):

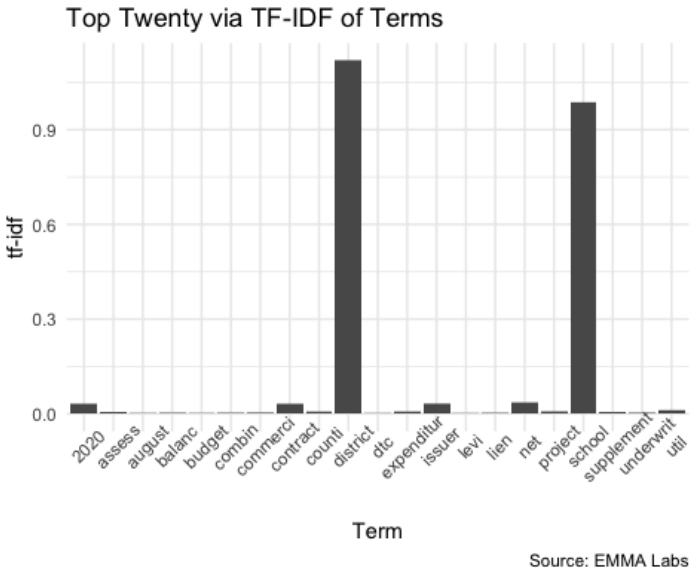
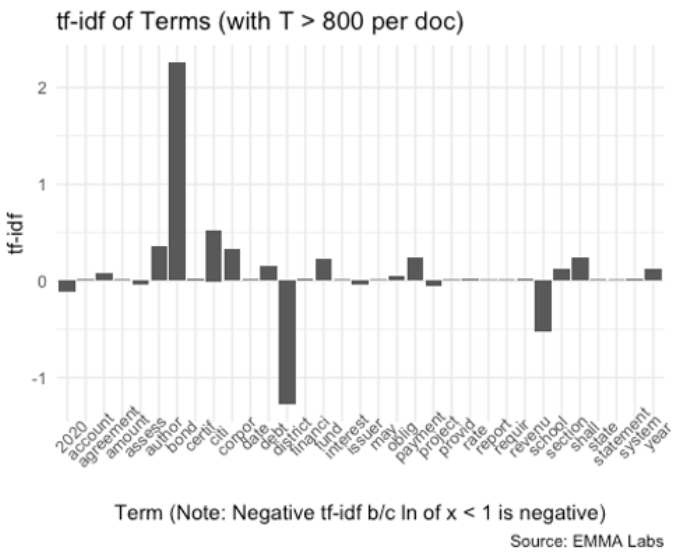


Figure 5

This is in good concordance with the frequency listings noted above. We also plotted the TF-IDF for those terms that appear more than eight-hundred times within a given document (Figure 6). Common financial stems like ‘assess’ and ‘certif’ dominate in our corpus:

Figure 6



We also measured which terms appeared most often together as a pair, and how often they did so (2017, 61-63). We can see from the graph of the top thirty most common pairs (Figure 7) that phrases involving ‘giving’, ‘receiving’ and ‘ownership’ predominate in the corpus. Perhaps indicating a focus on obligation, which would accord with the role of these documents as financial agreements.

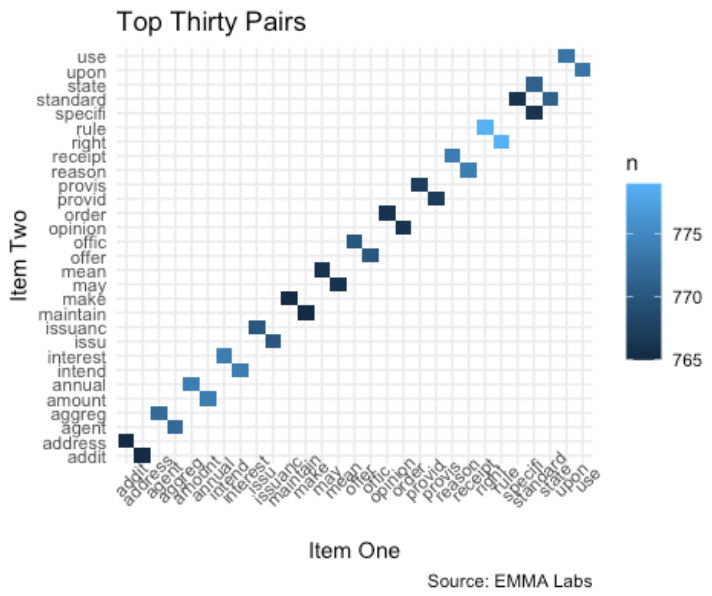


Figure 7

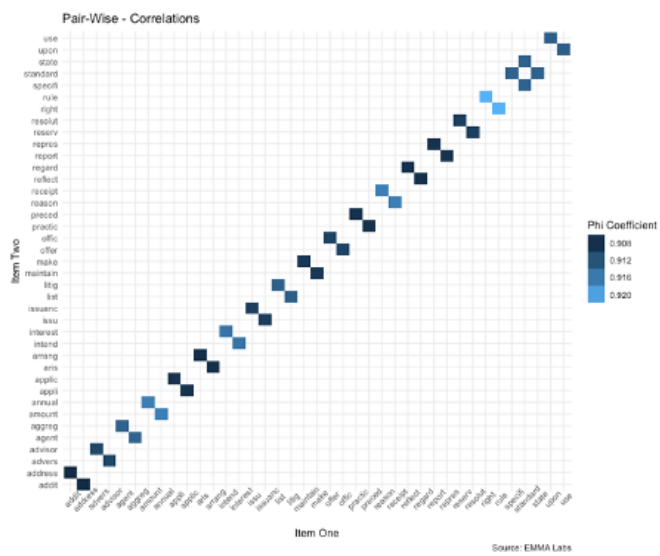


Figure 8

We also measured pairwise correlation between the top thirty paired-words in our corpus (Figure 8). This metric was built upon the 'phi coefficient', which measures how often two terms

are joined together rather than apart (2017, 63-66). Pairings such as ‘resolut-reserv’, ‘offer-make’, and ‘repres-regard’ dominate this metric.

We also plotted a network graph of those words that possess a phi-coefficient of more than .88 (2017, 56-66) with the ‘ggraph’ package (Pedersen 2024) (Figure 9):

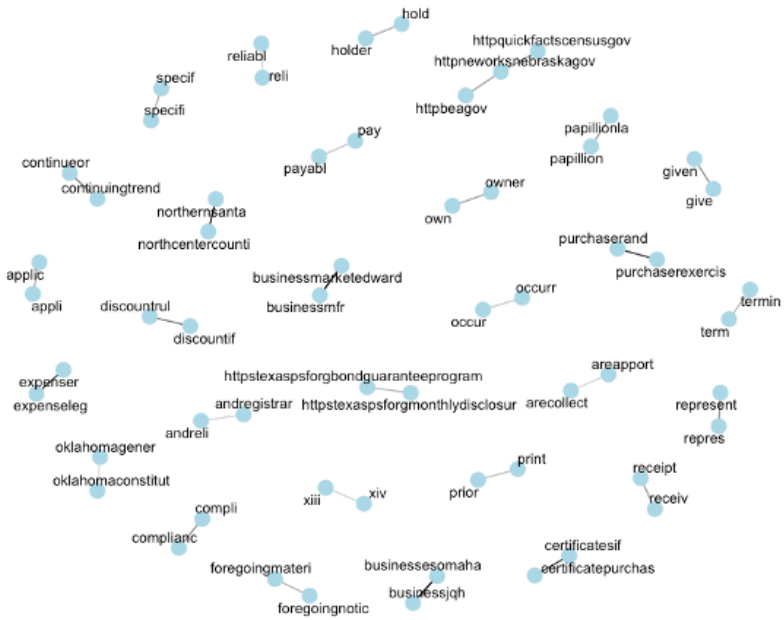


Figure 9

The occurrence terms like ‘pay’, ‘hold’, ‘business’ and so on reinforce the perception that our corpus is one with a financial/formal orientation.

In order to better gain a grasp of the relations of terms in our corpus, we also analyzed the two-word relationships, or bi-grams, that occurred throughout it (Figure 10 and Figure 11). To accomplish this, we made use of the ‘quanteda’, ‘dplyr’ packages (Hall 2021), and produced graphical output utilizing the ‘ggplot2’ (Silge and Robinson 2017, 56-60) and the ‘ggraph’ packages (Pedersen 2024).

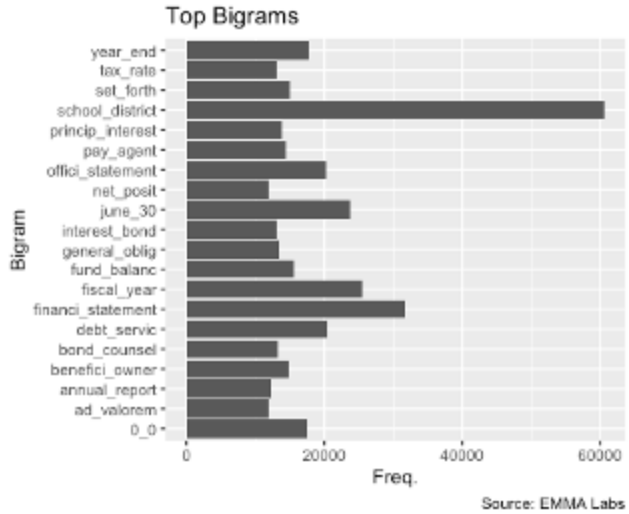


Figure 10

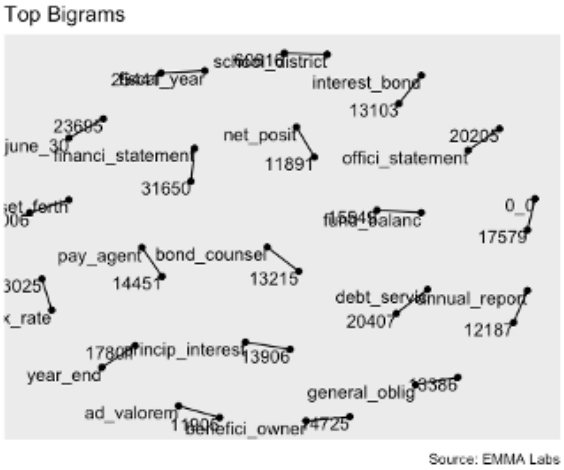


Figure 11

Note that the network graph in Figure 11 was built on an unfiltered DTM, so number pairings were included. These combinations, such as ‘fiscal-year’ and ‘pay-agent’ are also in concordance with what would be expected from standard financial bond documents.

An additional set of metrics with which to examine the structure of our corpus can be obtained via the property of lexical diversity (Figure 13). Using the *quanteda.textstats* library’s ‘textstat_readability’ function (Benioit et al. 2024b, 13-14), as well as the ‘ggpubr’ package (Kassambara n.d.), we were able to analyze our corpus via various metrics such as the Flesch's Reading Ease Score (Flesch), Gunning's Fog Index (FOG), Simple Measure of Gobbledygook (SMOG), and the Linsear Write measures (Figure 12).

The SMOG metric is used by researchers across a range of disciplines to measure the years of schooling required to understand a piece of text (Brown 2024). In our corpus, the most complicated documents by this measure require about 23 years of educational attainment – or, about a Phd’s worth of schooling. The average score for our entire corpus was 18.54 years, about a master’s degree worth. The Flesch metric is a standard measure of how easily a piece of text can be read (Gerstle 2010). A value of 100 indicates fairly easy comprehensibility (Izgi and

Seker 2012, 3). Our corpus averaged 26.542, indicating the relative complexity of texts that make up our corpus.

The FOG score is similar to the Flesch metric, with anything beyond '15' probably too difficult for most readers (Izgi and Seker 2012, 3). On this front, our corpus scored an average of 22.26. Finally, the Linsear Write metric is structured similarly to SMOG's, with a higher score indicating a higher education required to understand it (Power 2006). Our corpus averaged a score of 29.28 overall via this measure.

In sum, these measurements indicate that our corpus is made up of relatively sophisticated texts that do not ameen themselves to easy understanding.

```
   document            Flesch            FOG             SMOG        Linsear.Write
 Length:845         Min.   : 1.938   Min.   :16.08   Min.   :14.02   Min.   :14.54
 Class :character   1st Qu.:23.553   1st Qu.:21.28   1st Qu.:18.02   1st Qu.:27.16
 Mode  :character   Median :27.179   Median :21.89   Median :18.40   Median :28.58
                    Mean   :26.542   Mean   :22.26   Mean   :18.54   Mean   :29.28
                    3rd Qu.:29.100   3rd Qu.:23.16   3rd Qu.:19.10   3rd Qu.:31.27
                    Max.   :56.404   Max.   :31.26   Max.   :23.09   Max.   :48.83
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Figure 12

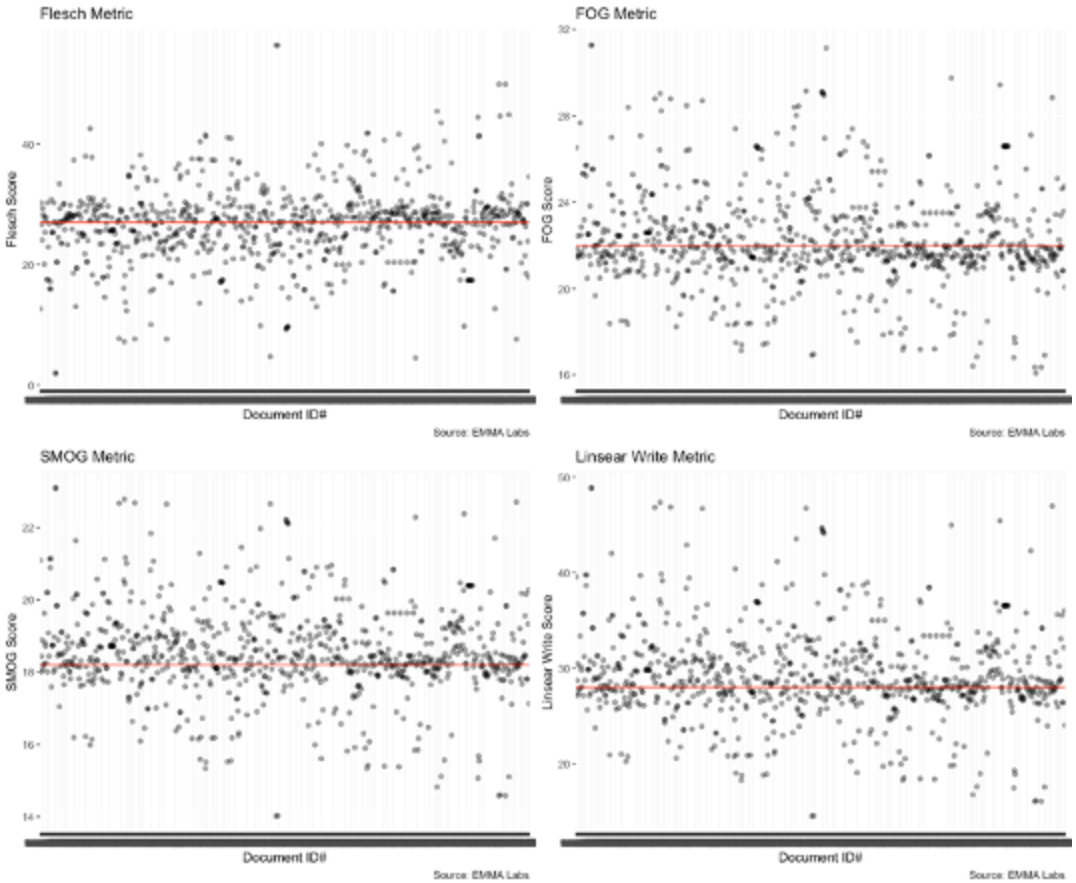


Figure 13

Dictionary Analysis

Our rule-based approach centered upon using dictionary methods to ascertain if our corpus could be construed as displaying an obvious political influence or propensity.

Our custom phrases used to construct our dictionary consisted of two broad categories composed of a total of twenty-seven words total. These were further broken down into words typically associated with governmental operations and those usually associated with political power-holding. The full dictionary is as follows:

Structural-Terms: 'mayor', 'budget', 'expenditures', 'city', 'revenue', 'administration', 'Tax', 'lobby', 'property', 'budget', 'bureaucracy', 'spending', 'cuts', 'funds'.

Power-Terms: 'partisanship', 'election', 'power', 'separation', 'federalism', 'fiscal', 'executive', 'reforms', 'constitution', 'district', 'officer', 'authority', 'governor', 'president'.

Using the ‘quanteda’ package (Beniot et al. 2024a, 34-36), we parsed our corpus using this dictionary. On average, texts in our corpus contained words from the former category 342.3 times, and words from the latter 271.5 times. This gives a *prima-facie* case for our corpus at least containing some words usually associated with some aspects of political dynamics within its constituent texts. Figures 14 and 15 demonstrate

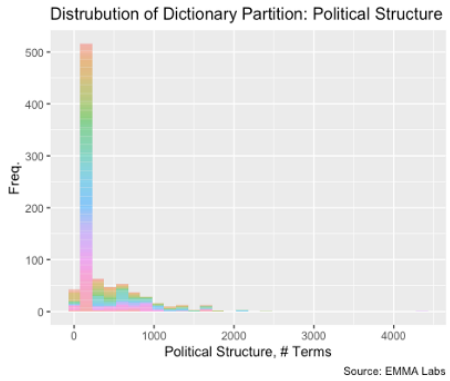


Figure 14

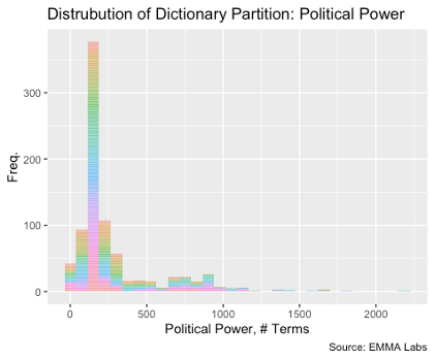


Figure 15

this corpus-level distribution of the categorical facet of our custom dictionary. Further, we can visualize the spread of these categories on a *per document* basis.

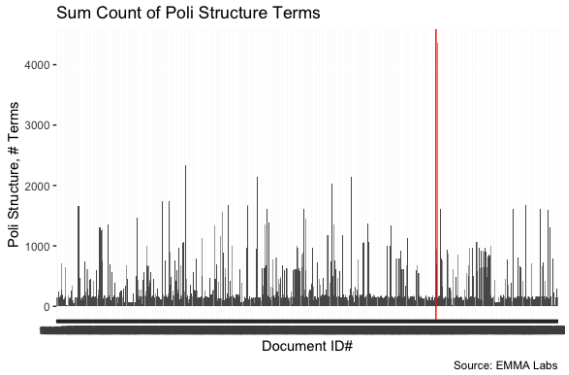


Figure 16

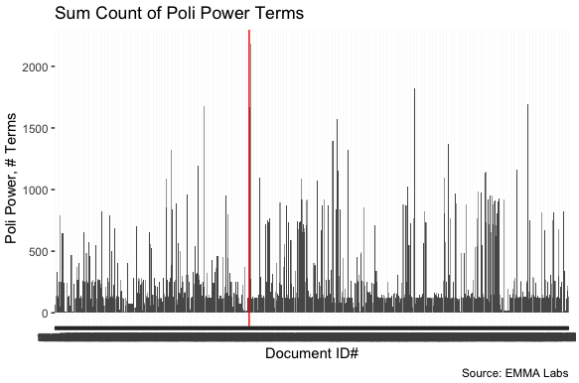
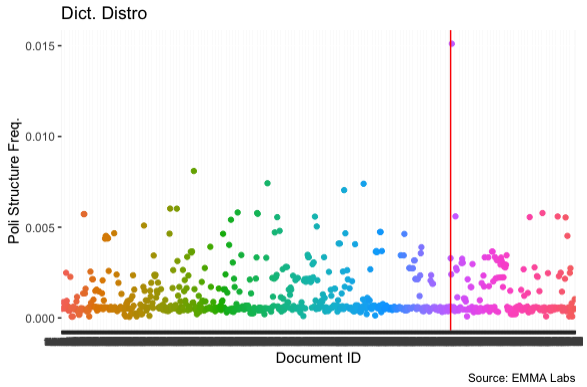


Figure 17



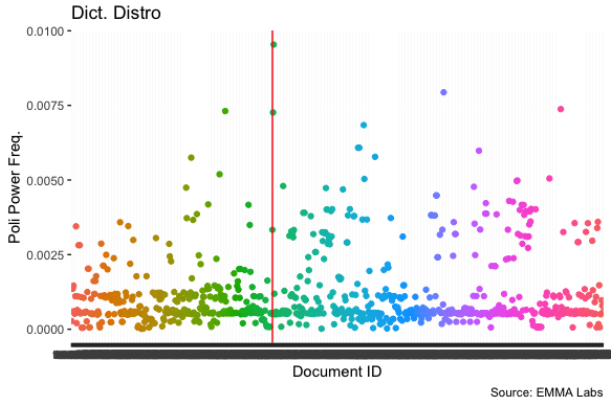


Figure 18 Figure 19

Figures 16 and 17 allow us to see that 'poli structure' terms are most common in documents around #640 in our corpus, and that documents #320 are host to the most frequent count of 'poli power' terms. Figures 18 and 19 present this information via the frequency of each category of dictionary terms per document. This information could be of use for further analysis, allowing future examinations to focus on those documents most likely to bear a stronger relationship to the underlying political sentiments we are trying to explore with our custom dictionary.

Unsupervised Approaches

Supervised methods, which rely on labeled data, are powerful for testing hypotheses and analyzing specific variables within a dataset. Comparing results from unsupervised methods such as Structural Topic Modeling (STM) may uncover latent patterns and associations in the data that do not align with pre-existing labels or hypotheses. Additionally, unsupervised methods validate or challenge the insights from supervised methods, ensuring findings are both reliable and comprehensive.

The combination of supervised and unsupervised findings provides a holistic view of the data. Supervised methods yield precise, hypothesis-driven insights, while unsupervised

approaches offer broader contextual understanding and the flexibility to identify unexpected patterns. Additionally, STM provide valuable insights into latent patterns within the text data without relying on predefined labels.Together, these methods enable us to validate results, explore new avenues, and generate well-rounded conclusions.

We conduct the unsupervised methods using the ‘stm’ package (Roberts et al. 2023). The data is processed in the same manner as in the supervised methods. First, the data is imported and explored to ensure familiarity with its structure. Next, text preprocessing is performed using the textProcessor function, which cleans and standardizes the text by converting it to lowercase, removing punctuation and stopwords, and stemming words to their root forms.

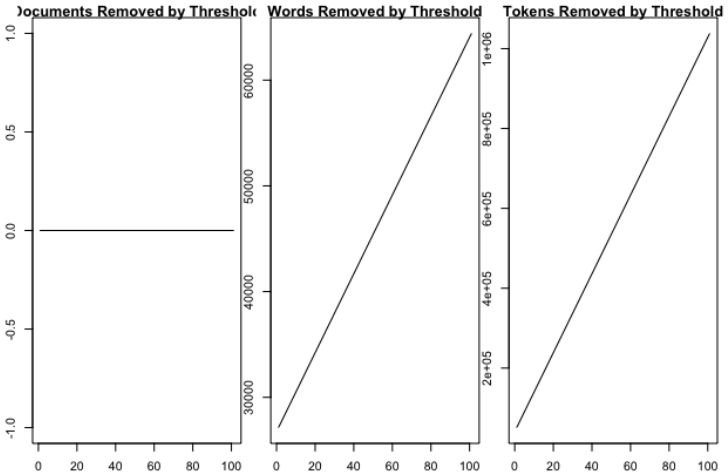
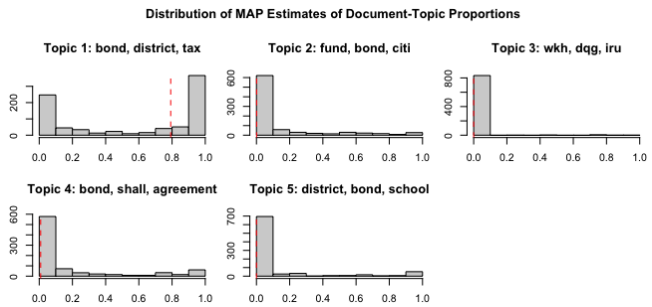


Figure 20

The cleaned data is then prepared for STM analysis using the prepDocuments function, which organizes the data into a document-term matrix and generates a vocabulary and metadata. The STM model is estimated using the stm function, specifying the number of topics (e.g., 10). The model’s output reveals the latent topics present in the dataset. Finally, various visualization tools, such as summary plots and topic correlation networks, are used to interpret the topics and

their relationships. This structured approach ensures that the insights derived from unsupervised analysis are both systematic and meaningful, complementing the findings from supervised methods.

In the following two visuals, we first notice that the topics that dominate the corpus are "bond issuance agreements" and "district specific funding mechanisms".



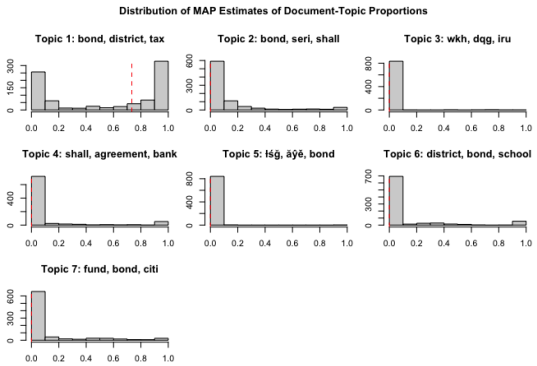


Figure 21 Figure 22

These topics are indicative of political risks arising from governance concerns in the municipal bond markets of the selected bonds.

The topic correlation plots shows that there is a weak correlation between the topics "district taxes" and "fund management". This may have occurred as these documents attempt to describe how taxes collected from specific districts are managed or allocated for municipal projects.

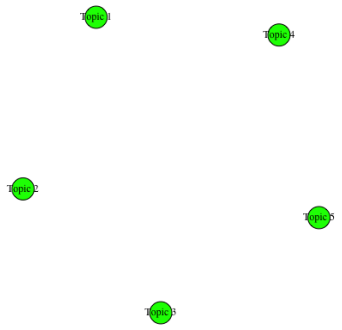


Figure 23

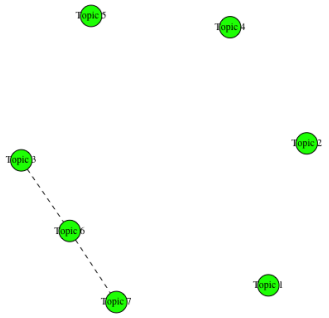


Figure 24

A few terms linked to political risks, such as: “tax”, “revenue” and “governor” are present across multiple topics. Some documents focused on administrative structures while others emphasized on various fiscal policy concerns.

The unsupervised analysis complements findings from the supervised approach, offering a broader perspective on how political risks are embedded within the text. The STM results underscore the complexity of these documents and highlight the diverse ways in which political and financial elements intersect. For instance, topics related to taxation and governance provide indirect evidence of political volatility impacting municipal bonds, as indicated by correlations with terms such as "district" and "revenue."

Conclusion

Ultimately, our study delves into the relationship between municipal bond offering documents and political risk through a comprehensive set of methods, leveraging a robust set of metrics and a large sample size to ensure our findings are representative. Within the analyzed documents, we identified frequently occurring terms such as “tax” and “bond.” While these terms appear adjacent to political risks, they do not directly disclose or signify local political

instability. Instead, their presence highlights broader thematic linkages that may be critical in understanding contextual underpinnings of political risk.

Our findings serve as a foundation for further exploration, suggesting key topics and patterns that warrant deeper investigation. At this time however, we fail to reject our null hypothesis; we conclude that political risk is not directly disclosed within bond offering documents. This study underscores the potential of examining textual elements in municipal bond documents as a lens for understanding political risk, offering a pathway for future studies to build upon and refine these initial insights.

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